

Department of Nano Science Technology

Faculty of Technology

Wayamba University of Sri Lanka

Centralized Workshops and Laboratory Allocation Platform

UI Implementation and Verification

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Module: NANO2142

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1. Executive Summary & Purpose of the Verification Document

This technical report documents the end-to-end user interface implementation and system verification for the SYNCLAB Centralized Workshop & Laboratory Allocation Platform. By incorporating actual runtime interface captures, this document validates the structural fidelity of the client-side data entry pipeline, defensive input validation modals, cryptographic key generation upon reservation confirmation, and comprehensive administrative search, filtering, and reporting tools.

Quality & Compliance Objectives

• Operational Data Entry: Validate seamless user input capture across mandatory text fields, dropdown selectors, and extended notes.
• Defensive Validation & User Feedback: Confirm that erroneous, malformed, or omitted inputs trigger unambiguous modal dialogs with focus retention.
• Transactional State Persistence: Demonstrate issuance of unique 4-digit allocation keys to support zero-trust modification without account overhead.
• Administrative Governance & Audit Reporting: Verify multi-parameter date filtering, real-time timetable integration, and tabular data export.

2. Data Entry Interface: Finalize Your Allocation

The Data Entry interface ('Finalize your allocation') represents the final user stage in the self-service facility reservation process. It binds student credentials, department affiliation, supervisor mentorship, and research project metadata to an actively selected venue and pre-locked calendar slot. The interface maintains visual hierarchy through an allocation context banner, structured input fields with individual reset controls, and clear action triggers.

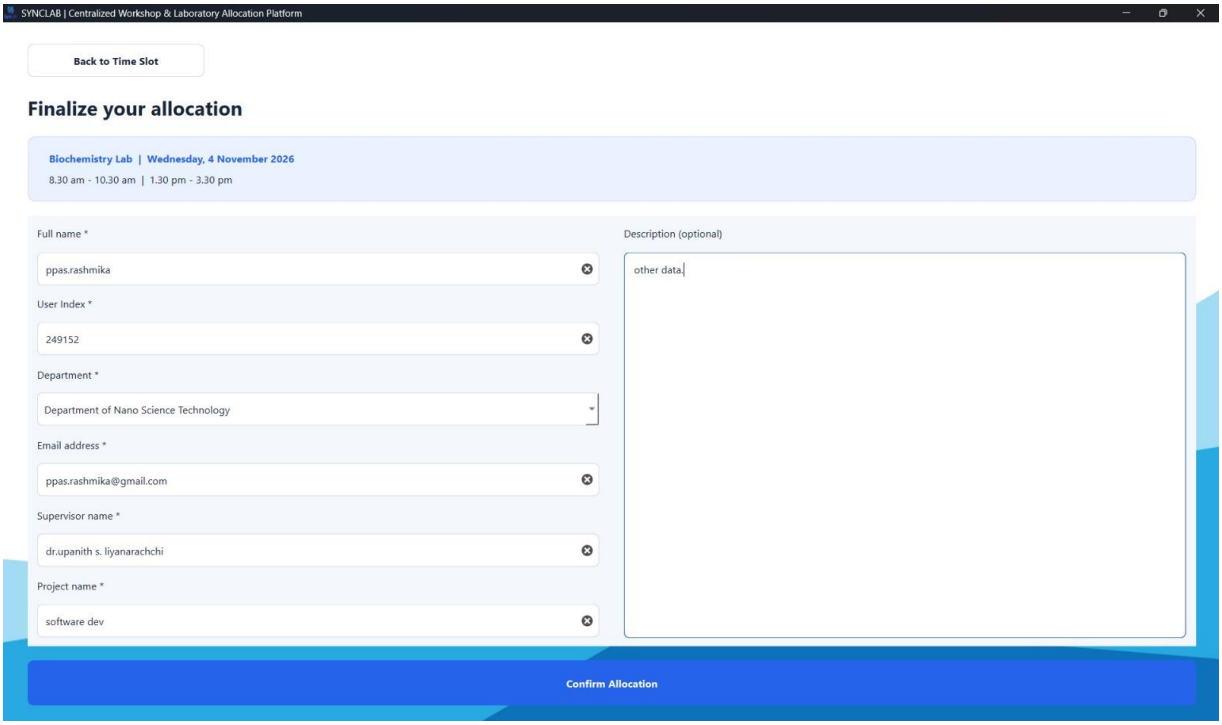


Figure 2.1: SYNCLAB 'Finalize your allocation' Data Entry Interface with Captured Field Values

Field-by-Field Input Specification & Runtime State

- **Facility Context Banner:** Displays locked venue ('Biochemistry Lab'), target date ('Wednesday, 4 November 2026'), and scheduled slot intervals ('8.30 am - 10.30 am | 1.30 pm - 3.30 pm').
- **Full Name (*):** Text input for applicant identification ('ppas.rashmika') equipped with dynamic clear button ('✕').
- **User Index (*):** Numerical student index number ('249152') validated against campus registry records.
- **Department (*):** Constrained dropdown menu mapping to academic faculties ('Department of Nano Science Technology').
- **Email Address (*):** Institutional contact address ('ppas.rashmika@gmail.com') bound to regex pattern verification.
- **Supervisor Name (*):** Direct research advisor endorsement ('dr.upanith s. liyanarachchi').
- **Project Name (*):** Formal experimental title or study module ('software dev').
- **Description (Optional):** Multi-line text area ('other data.') providing additional equipment specifications or notes.

3. Input Validation & Error Handling Dialogs

To ensure database integrity and enforce academic governance, the SYNCLAB platform executes synchronous validation prior to form submission. If any input fails format criteria or omits mandatory parameters, execution is halted, preserving user input state while presenting an unambiguous modal challenge that directs corrective action.

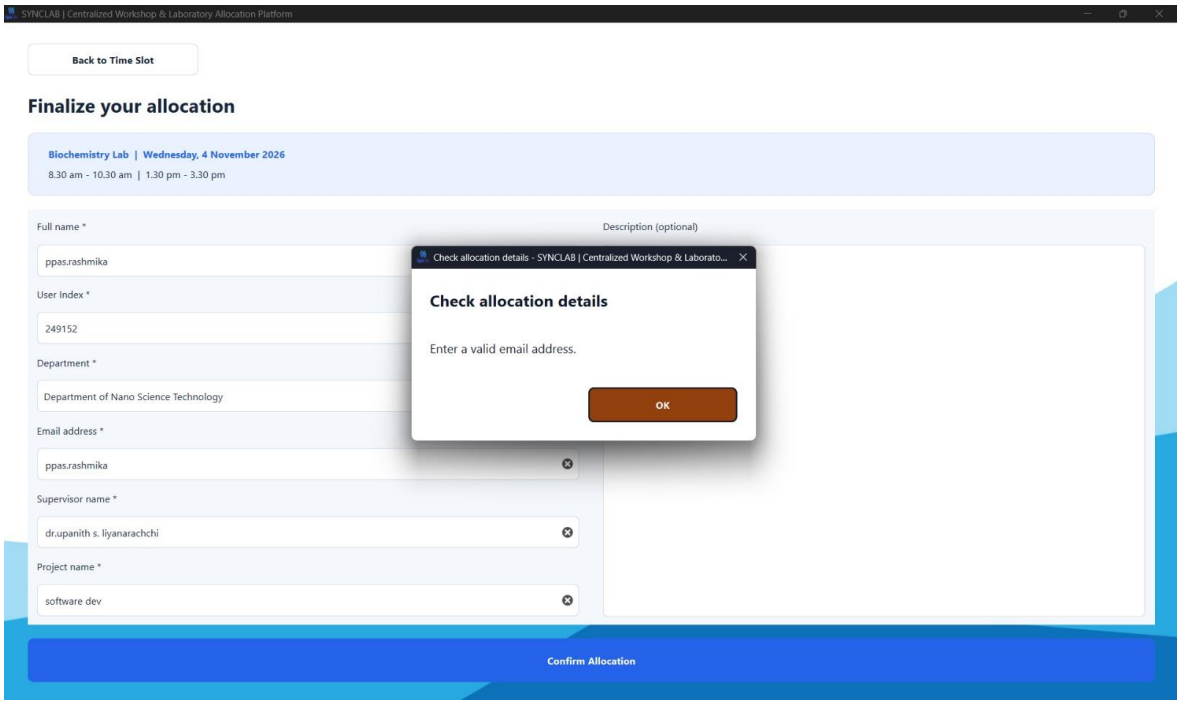


Figure 3.1: Runtime Validation Modal – Email Syntax Format Error ('Enter a valid email address.')

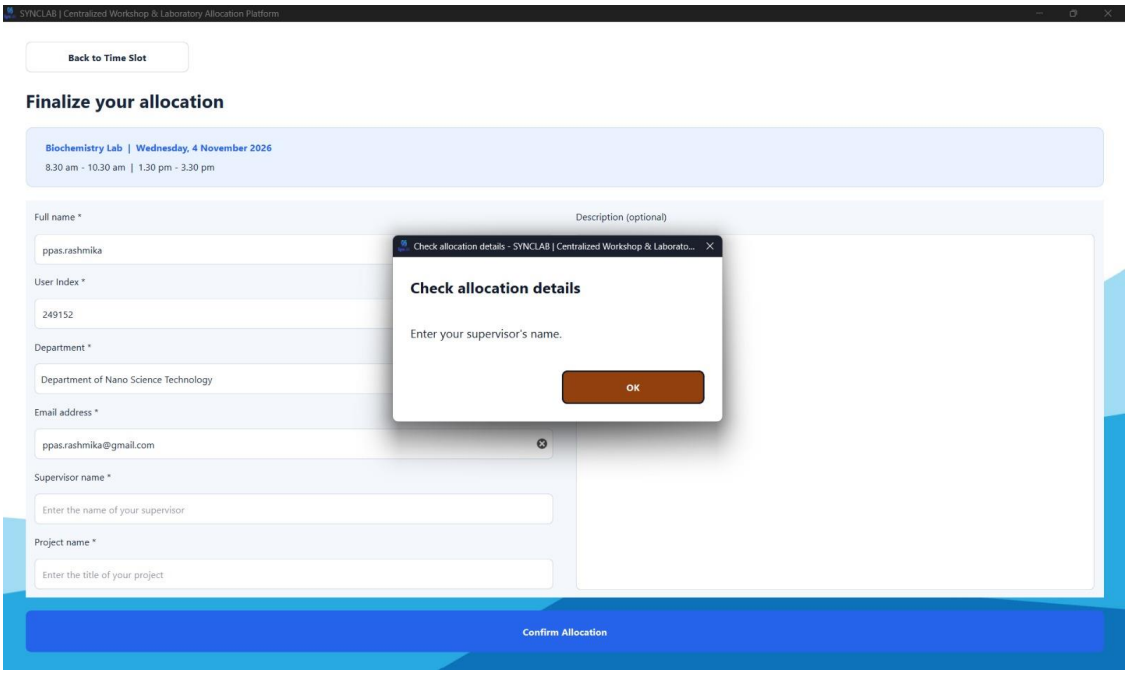


Figure 3.2: Runtime Validation Modal – Mandatory Field Omission ('Enter your supervisor\'s name.')

Validation Logic & Error Trapping Matrix

Form Input Field	Validation Type	Triggering Condition	Observed Modal Feedback
Email Address	Regex Pattern Matching	User enters string without '@domain.com' (e.g., 'ppas.rashmika')	Modal Alert: 'Enter a valid email address.'
Supervisor Name	Mandatory Presence Check	User leaves supervisor input field completely blank	Modal Alert: 'Enter your supervisor\'s name.'
User Index	Format & Numeric Check	User enters non-numeric or out-of-range index number	Modal Alert: 'Enter a valid user index number.'
Project Name	Mandatory Presence Check	User omits research project or module designation	Modal Alert: 'Enter the title of your project.'

4. Successful Allocation & Cryptographic Key Generation

Upon successful validation and conflict verification against concurrent database locks, the platform commits the record to the active repository. A modal confirmation displays the reservation summary and generates an authentic 4-digit numeric allocation key ('1819'). This token grants zero-trust editing and cancellation rights to the student without requiring permanent user accounts.

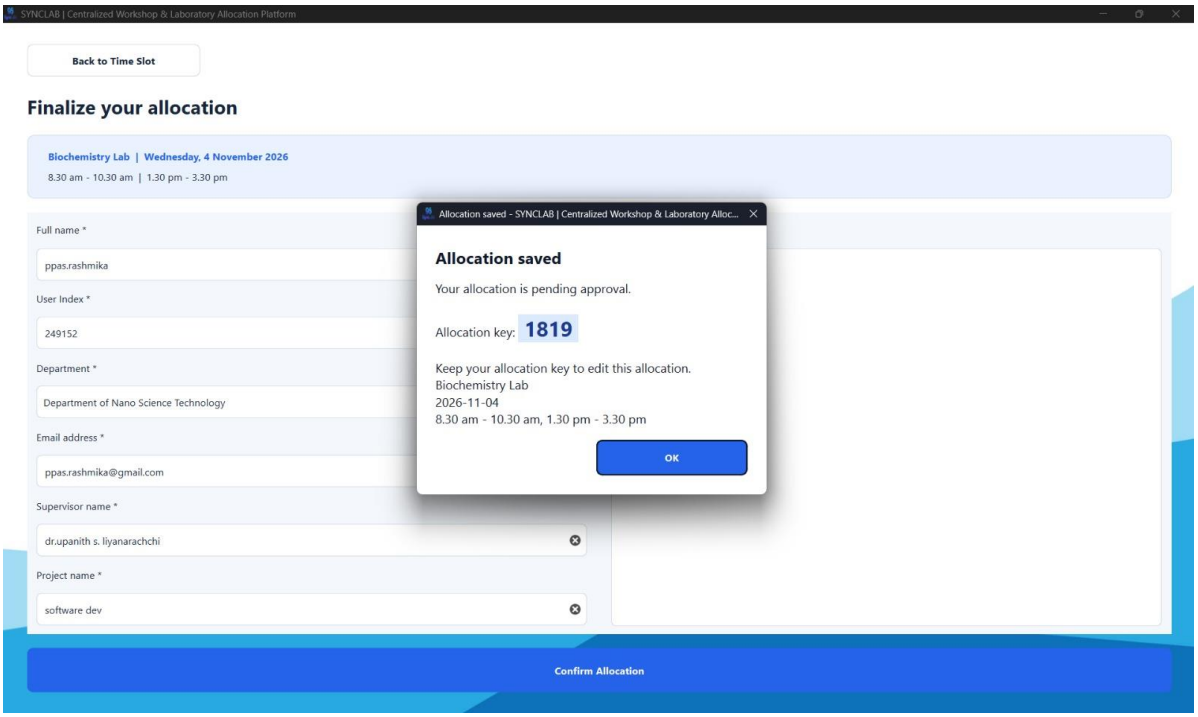


Figure 4.1: Reservation Success Modal Displaying Generated Allocation Key: 1819 & Session Summary

Transaction Confirmation Attributes

- **Workflow Status:** Marked as 'Your allocation is pending approval', signaling routing to lab coordinator queue.
- **Issued Allocation Key:** Unique token '1819' issued to student for subsequent record modifications or revocations.
- **Bound Session Details:** Confirmed for 'Biochemistry Lab' on '2026-11-04' across time intervals '8.30 am - 10.30 am, 1.30 pm - 3.30 pm'.
- **Navigation Dismissal:** Clicking '[OK]' safely resets the reservation wizard state and redirects to the landing dashboard.

5. Administrative Dashboard: Search, Filter, and Master Reporting

The SYNCLAB Administrator Interface provides facility managers with comprehensive oversight of laboratory utilization. The screen integrates an interactive calendar for single-day filtering, quick status toggles ('Edit', 'Delete', 'Pending', 'Rejection'), a global toggle ('Show All Dates'), and an audit-ready tabular report listing active reservations with full applicant metadata.

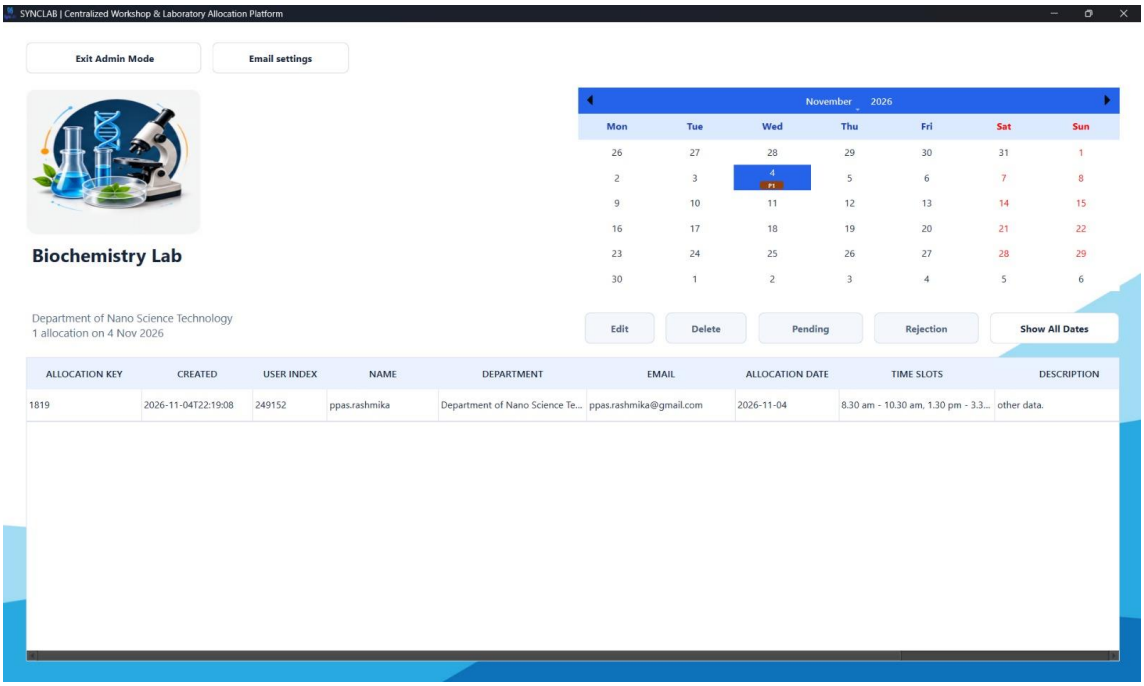


Figure 5.1: Administrator Dashboard Featuring Interactive Calendar, Filter Controls, and Real-Time Allocation Audit Grid

Search & Filter Interactive Capabilities

- **Calendar-Based Filtering:** Selecting 'Wed 4 Nov 2026' isolates reservations scheduled for that operational date (designated with count badge 'P1').
- **Global Scope Toggle ('Show All Dates'):** Clears single-day date constraints to populate the comprehensive multi-week laboratory schedule.
- **Status Action Buttons:** Direct access buttons to toggle record sets between '[Pending]', '[Rejection]', '[Edit]', and '[Delete]'.
- **Top Utility Controls:** '[Exit Admin Mode]' securely terminates elevated privileges; '[Email settings]' configures automated notification triggers.

Administrative Allocation Report & Audit Schema

The runtime allocation report generated in the bottom panel reflects the exact live data committed during testing:

Key	Created Timestamp	User Index	Name	Department	Email Address	Alloc Date	Time Slots	Description
1819	2026-11-04T22:19:08	249152	ppas.rashmika	Department of Nano Science Technology	ppas.rashmika@gmail.com	2026-11-04	8.30 am - 10.30 am, 1.30 pm - 3.30 pm	other data.